CIPROFLO 2MG/ML SOLUTION FOR IV INFUSION
Ciprofloxacin 2mg/ml Solution for IV infusion

Description
A colourless to slightly yellow clear sterile solution in glass vial.

Composition
Each 1 ml contains 2mg of Ciprofloxacin.

Summary of pharmacodynamics and pharmacokinetics
Ciprofloxacin is a synthetic 4-quinolone derivative, with bactericidal activity. It acts via inhibition of bacterial DNA gyrase, ultimately resulting in interference with DNA function. Ciprofloxacin is highly active against a wide range of Gram-positive and Gram-negative organisms and has shown activity against some anaerobes, *Chlamydia* spp. and *Mycoplasma* spp.. Killing curves demonstrate the rapid bactericidal effect against sensitive organisms and it is often found that minimum bactericidal concentrations are in the range of minimum inhibitory concentrations. Ciprofloxacin has been shown to have no activity against *Treponema pallidum* and *Ureaplasma urealyticum, Nocardia asteroides,* and *Enterococcus faecium* are resistant.

Susceptibility
The prevalence of resistance may vary geographically and with time for selected species and local area information on resistance is desirable, particularly when treating severe infections. This information gives only an approximate guidance on probabilities whether micro-organisms will be susceptible to ciprofloxacin or not.

Plasmid-related transfer of resistance has not been observed with ciprofloxacin and the overall frequency of development of resistance is low (10^{-9} - 10^{-7}). Cross-resistance to penicillins, cephalosporins, aminoglycosides and tetracyclines has not been observed and organisms resistant to these antibiotics are generally sensitive to ciprofloxacin. Ciprofloxacin is also suitable for use in combination with these antibiotics, and additive behavior is usually observed.

Absorption of oral doses of ciprofloxacin 250mg, 500mg and 750mg tablet formulation occurs rapidly, mainly from the small intestine, the half-life of absorption being 2-15 minutes. Plasma levels are dose-related and peak 0.5-2.0 hours after oral dosing. The AUC also increases dose proportionately after administration of both single and repeated oral (tablet) and intravenous doses. The pharmacokinetic profile of intravenous ciprofloxacin was shown to be linear over the dose range (100mg-400mg). Following intravenous administration of ciprofloxacin, the mean maximum plasma concentrations were achieved at the end of the infusion period. That is, for a 100mg or 200mg dose, 30 minutes, and for a 400mg dose, 60 minutes. Reported plasma levels at this time point were 1.8mg/l, 3.4mg/l and 3.9mg/l, respectively. The absolute bioavailability is reported to be 52-83% and ciprofloxacin is subject to only slight first-pass metabolism.

Distribution of ciprofloxacin within tissues is wide and the volume of distribution high, though slightly lower in the elderly. Protein binding is low (between 19-40%). Ciprofloxacin is present in plasma largely in a non-ionised form.

Only 10-20% of a single oral or intravenous dose is eliminated as metabolites (which exhibit lower activity than the parent drug). Four different antimicrobially active metabolites have been reported, desethyleneciprofloxacin (M1), sulphociprofloxacin (M2), oxaciprofloxacin (M3) and formylciprofloxacin (M4). M2 and M3 account for one third each of metabolised substance and M1 is found in small amounts (1.3-2.6% of the dose). M4 has been found in very small quantities (<0.1% of the dose). M1-M3 have antimicrobial activity comparable to nalidixic acid and M4 found in the smallest quantity has antimicrobial activity similar to that of norfloxacin.

Elimination of ciprofloxacin and its metabolites occurs rapidly, primarily by the kidney. After single oral and intravenous doses of ciprofloxacin, 55% and 75% respectively are eliminated by the kidney and 39% and 14% in the faeces within 5 days. Renal elimination takes place mainly during the first 12 hours after dosing and renal clearance levels suggest that active secretion by the renal tubules occurs in addition to normal glomerular filtration. Renal clearance is between 0.18-0.3 l/h.kg and total body clearance between 0.48-0.60 l/h.kg. The elimination kinetics are linear and after repeated dosing at 12 hourly intervals, no further accumulation is detected after the distribution equilibrium is attained (at 4-5 half-lives). The elimination half-life of unchanged ciprofloxacin over a period of 24-48 hours post-dose is 3.1-5.1 hours. A total body clearance of approximately 35l/h was observed after intravenous administration.

Some studies carried out with ciprofloxacin in severely renally impaired patients (serum creatinine>265 micromole/l or creatinine clearance <20ml/minute) demonstrated either a doubling of the elimination half-life, or fluctuations in half-life in comparison with healthy volunteers, whereas other studies showed no significant correlation between elimination half-life and creatinine clearance. However, it is recommended that in severely renally impaired patients, the total daily dose should be reduced by half, although monitoring of drug serum levels provides the most reliable basis for dose adjustment as necessary.

Results of pharmacokinetic studies in paediatric cystic fibrosis patients have shown dosages of 20mg/kg orally twice daily or 10mg/kg iv three times daily are recommended to achieve plasma concentration/time profiles comparable to those achieved in the adult population at the currently recommended dosage regimen.

Indications

Ciproflo 2mg/ml solution for infusion is indicated in the treatments of uncomplicated and complicated infections caused by ciprofloxacin sensitive pathogens:

- Infections of the respiratory tract
  In the treatment of outpatients with pneumonia due to Pneumococcus ciprofloxacin should not be used as a first choice of drug. Ciprofloxacin can be regarded as an advisable treatment for pneumonias caused by Klebsiella, Enterobacter, Proteus, E. coli, Pseudomonas, Haemophilus, Branhamella, Legionella, and Staphylococcus.
- Infections of the middle ear (otitis media), of the paranasal sinuses (sinusitis), especially if these are caused by gram negative organisms including Pseudomonas or by Staphylococcus.
- Infections of the eyes
- Infections of the kidneys and/or the efferent urinary tract
- Infections of the genital organs, including adnexitis, gonorrhoea, prostatitis
- Infections of the abdominal cavity (e.g. infections of the gastrointestinal tract or of the biliary tract, peritonitis)
- Infections of the skin and soft tissue
- Infections of the bones and joints
- Sepsis
- Infections or imminent risk of infection (prophylaxis) in patients whose immune system has been weakened (e.g. patients on immunosuppressants or have neutropenia)
- Selective intestinal decontamination in immunosuppressed patients

Adverse reactions/side effects

The most frequently reported adverse reactions are: nausea, diarrhoea and rash.
Adverse reactions are listed by frequency: common > 1/100, <1/10); uncommon > 1/1,000, < 1/100); rare > 1/10,000, < 1/1,000); very rare (< 1/10,000). The following adverse reactions have been observed:
Infections and infestations
Uncommon: moniliasis
Rare: moniliasis (oral), vaginal moniliasis
Very rare: moniliasis (gastrointestinal)

Blood and the lymphatic system disorders
Uncommon: eosinophilia, leukopenia
Rare: anaemia, leukopenia (granulocytopenia), leucocytosis, thrombocytopenia, thrombocythaemia (thrombocytosis)
Very rare: hemolytic anaemia, agranulocytosis, pancytopenia (life ‒ threatening), bone marrow depression (life ‒ threatening)

Immune system disorders
Rare: allergic reaction, anaphylactoid (anaphylactic) reaction, dyspnoea, larynx oedema.
Very rare: shock (anaphylactic/anaphylactoid reactions progressing in very rare cases to life ‒ threatening shock), serum sickness like reaction, angioedema, pruritic rash

Metabolism and nutrition disorders
Uncommon: anorexia

Rare: hyperglycaemia

Psychiatric and nervous system disorders
Uncommon: headache, dizziness, insomnia, agitation, confusion, taste perversion (usually reversible upon discontinuation of treatment)
Rare: hallucination, paresthesia (peripheral paralgesia), abnormal dreams (nightmares), depression, tremor, convulsion, hypesthesia, somnolence, exacerbation of symptoms of myasthenia gravis
Very rare: grand mal convulsion, abnormal gait, psychosis (which may progress to self-endangering behaviour), intracranial hypertension, ataxia, hyperesthesia, hypertonia, parosmia (impaired smell), anosmia (usually reversible on discontinuation), migraine, anxiety, taste loss (impaired taste)

Eye disorders
Very rare: abnormal vision (visual disturbances), diplopia, chromatopsia

Ear and labyrinth disorders
Very rare: tinnitus, transitory deafness (especially at high frequencies)

Cardiac and vascular disorders
Rare: tachycardia, syncope (fainting), vasodilation (hot flushes), hypotension
Very rare: vasculitis (petechiae, haemorrhagic bullae, papules, crust formation)

Gastrointestinal disorders
Common: nausea, diarrhoea
Uncommon: vomiting, dyspepsia, flatulence, abdominal pain
Rare: dysphagia, pseudomembranous colitis, pancreatitis
Very rare: life‑threatening pseudomembranous colitis with possible fatal outcome
Hepato-biliary disorders
Uncommon: bilirubinaemia,
Rare: jaundice, cholestatic jaundice, liver necrosis (rarely progressing to life‑threatening hepatic failure)
Very rare: hepatitis

Skin and subcutaneous tissue disorders
Common: rash
Uncommon: pruritis, maculopapular rash, urticaria
Rare: photosensitivity reaction, erythema mutliforme (minor), erythema nodosum
Very rare: Stevens-Johnson-Syndrome, epidermal necrolysis (Lyell-Syndrome), fixed drug reaction, petechia

Musculoskeletal, connective tissue and bone disorders
Uncommon: arthralgia
Rare: myalgia, joint disorder (joint swelling), pain in extremeties, back pain, myasthenia
Very rare: twitching, tendinitis (predominantly achillo tendinitis including tenosynovitis), partial or complete tendon rupture (predominantly achilles tendon)

Renal and urinary disorders
Rare: acute kidney failure, abnormal kidney function, haematuria, crystalluria, interstitial nephritis

General disorders and administration site conditions
Uncommon: asthenia (general feeling of weakness, tiredness), injection site reaction (e.g. oedema/ hypersensitivity/ inflammation/ pain), (thrombo)-phlebitis (at infusion site)
Rare: pain, chest pain, sweating, drug fever, oedema (peripheral, vascular, face)

Investigations
Uncommon: SGOT increased, SGPT increased, abnormal liver function test, alkaline phosphatase increased, increases in creatinine, increases in BUN (urea)
Rare: altered prothrombin values
Very rare: amylase increased, lipase increased

Post Marketing Experience
Exacerbation of myasthenia gravis

Precautions/warnings

In the event of hypersensitivity, which in some instances can occur after the first administration, therapy should be discontinued.
Ciprofloxacin should be used with caution in epileptics and patients with a history of CNS disorders and only if the benefits of treatment are considered to outweigh the risk of possible CNS side-effects. CNS side-effects have been reported after first administration of ciprofloxacin in some patients. Treatment should be discontinued if the side-effects, depression or psychoses lead to self-endangering behaviour).
Crystalluria related to the use of ciprofloxacin has been reported. Patients receiving ciprofloxacin should be well hydrated and excessive alkalinity of the urine should be avoided.
Patients with a family history of or actual defects in glucose-6-phosphate dehydrogenase activity are prone to haemolytic reactions with quinolones, and so ciprofloxacin should be used with caution in these patients.
Tendon inflammation and rupture may occur with quinolone antibiotics. Such reactions have been observed particularly in older patients and in those treated concurrently with corticosteroids. At the first sign of pain or inflammation, patients should discontinue ciprofloxacin and rest the affected limbs.
There is a risk of pseudomembranous colitis with broad-spectrum antibiotics possibly leading to a fatal outcome. It is important to consider this in patients suffering from severe, persistent diarrhoea. With ciprofloxacin this effect has been reported rarely. If pseudomembranous colitis is suspected treatment with ciprofloxacin should be stopped and appropriate treatment given (e.g. oral vancomycin). Drugs that inhibit peristalsis must not be given.
Ciprofloxacin has been shown to produce photosensitivity reactions. Patients taking ciprofloxacin should avoid direct exposure to excessive sunlight or UV-light. Therapy should be discontinued if photosensitisation (i.e., sunburn-like skin reactions) occurs.
Laboratory tests may give abnormal findings if performed whilst patients are receiving ciprofloxacin, e.g. increased alkaline phosphatase; increases in liver function tests, e.g. transaminases and cholestatic jaundice, especially in patients with previous liver damage.
In patients for whom sodium intake is of medical concern (e.g. patients with congestive heart failure, renal failure, nephrotic syndrome), the sodium content of Ciproflo Infusion should be taken into account.

<u>Special warning and precaution for use:</u>
Musculo-skeletal system:
The risk of developing fluoroquinolone-associated tendinitis and tendon rupture is further increased in people older than 60, in those taking corticosteroid drugs, and in kidney, heart, and lung transplant recipients. Patients experiencing pain, swelling, inflammation of a tendon or tendon rupture should be advised to stop taking their fluoroquinolone medication (Ciprofloxacin) and to contact their health care professional promptly about changing their antimicrobial therapy. Patients should also avoid exercise and using the affected area at the first sign of tendon pain, swelling, or inflammation.

Exacerbation of myasthenia gravis
Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Contraindications
Ciprofloxacin is contra-indicated in patients who have shown hypersensitivity to ciprofloxacin or any of the excipients, or other quinolone anti-infectives.
Except in cases of exacerbation of cystic fibrosis associated with *P. aeruginosa* (in patients aged 5- 17 years) and inhalation anthrax, ciprofloxacin is contra-indicated in children and growing adolescents unless the benefits of treatment are considered to outweigh the risks.
Concurrent administration of ciprofloxacin and tizanidine is contraindicated since an undesirable increase in serum tizanidine concentrations associated with clinically relevant tizanidine-induced side-effects (hypotension, somnolence) can occur.

Dosage and directions for use
Ciproflo 2mg/ml Solution infusions may be administered by intravenous infusion.

Posology and method of administration
Unless otherwise prescribed, the following guideline doses are recommended:

Intravenous
Respiratory tract infection
(according to severity and organism) 2 x 200-400 mg
Urinary tract infections:
- acute, uncomplicated 2 x 100 mg
- cystitis in women (before menopause) single dose 100 mg
- complicated 2 x 200 mg
Gonorrhea
- extragenital 2 x 100 mg
- acute, uncomplicated single dose 100 mg
Diarrhea 2 x 200 mg
Other infections (see indications) 2 x 200-400 mg
Particularly severe, life threatening 3 x 400 mg
Infections, i.e.
- Streptococcal pneumonia
- Recurrent infections in cystic fibrosis
- Bone and joint infections
- Septicemia
- Peritonitis
In particular when Pseudomonas, Staphylococcus or Streptococcus is present

After intravenous administration the treatment can be continued orally.

Incompatibilities
The ciprofloxacin infusion solution is compatible with physiological saline, Ringer solution and Ringer lactate solution, 5% and 10% glucose solutions, 10% fructose solution, and 5% glucose solution with 0.225% NaCl or 0.45% NaCl. When ciprofloxacin infusion solutions are mixed with compatible infusion solutions, for microbiological reasons and light sensitivity these solutions should be administered shortly after admixture.

Unless compatibility with other infusion solution/drugs has been confirmed, the infusion solution must always be administered separately. The visual signs of incompatibility are e.g. precipitation, clouding, and discoloration.

Incompatibility appears with all infusion solutions/drugs that are physically or chemically unstable at the pH of the solution adjusted to an alkaline pH (pH of the Ciprofloxacin infusion solutions : 3.9 - 4.5).

Duration of treatment
The duration of treatment depends on the severity of the illness and on the clinical and bacteriological course. It is essential to continue therapy for at least3 days after disappearance of the fever or of the clinical symptoms. Mean duration of treatment:
- 1 day for acute uncomplicated gonorrhoea and cystitis,
- up to 7 days for infections of the kidneys, urinary tract, and abnormal cavity.
- Over the entire period of the neutropenic phase in patients with weakened body defences,
- A maximum of 2 months in osteomyelitis,
- And 7 - 14 days in all other infections.

In streptococcal infections the treatment must last at least 10 days because of the risk of late complications. Infections caused by Chlamydia should also be treated for a minimum of 10 days.

Elderly
Elderly patients should receive a dose as low as possible depending on the severity of their illness and the creatinine clearance.

Children: contraindicated
Renal and hepatic impairment
1. Impaired renal function
1.1 Where creatinine clearance is between 31 and 60 ml/min/1.73m2 or where the serum creatinine concentration is between 1.4 and 1.9 mg/100ml the maximum daily dose should be 800mg per day for an intravenous regimen.

1.2 Where creatinine clearance is equal or is less than 30ml/min/1.73m2 or where the serum creatinine concentration is equal or higher than 2.0mg/100 ml the maximum daily dose should be 400 mg per day for an intravenous regimen.

2. Impaired renal function + haemodialysis
Dose as in 1.2; on dialysis days after dialysis

3. Impaired renal function + CAPD
Addition of ciprofloxacin infusion solution to the dialysate (intraperitoneal): 50mg ciprofloxacin/liter dialysate administered 4 times a day every 6 hours

4. Impaired liver function
No dose adjustment is required

5. Impaired renal and liver function
Dose adjustment as in 1.1 and 1.2

Use in pregnancy and lactation
Reproduction studies performed in mice, rats and rabbits using parenteral and oral administration did not reveal any evidence of teratogenicity, impairment of fertility or impairment of peri-/post-natal development. However, as with other quinolones, ciprofloxacin has been shown to cause arthropathy in immature animals, and therefore its use during pregnancy is not recommended. Studies have indicated that ciprofloxacin is secreted in breast milk. Administration to nursing mothers is thus not recommended.

Drug interactions
Increased plasma levels of theophylline have been observed following concurrent administration with ciprofloxacin. It is recommended that the dose of theophylline should be reduced and plasma levels of theophylline monitored. The reaction between theophylline and ciprofloxacin is potentially life-threatening. Therefore, where monitoring of plasma levels is not possible, the use of ciprofloxacin should be avoided in patients receiving theophylline. Particular caution is advised in those patients with convulsive disorders.
Ciprofloxacin inhibits CYP1A2 and thus may cause increased serum concentration of concomitantly administered substances metabolised by this enzyme (e.g. theophylline, clozapine, tacrine, ropinirole, tizanidine, duloxetine). Therefore, patients taking these substances concomitantly with ciprofloxacin should be monitored closely for clinical signs of overdose, and determination of serum concentrations, especially of theophylline, may be necessary.
Tizanidine must not be administered together with ciprofloxacin.
Concomitant use of duloxetine with strong inhibitors of the CYP450 1A2 isozyme such as fluvoxamine, may result in an increase of AUC and Cmax of duloxetine. Although no clinical data are available on a possible interaction with ciprofloxacin, similar effects can be expected upon concomitant administration.
Phenytoin levels may be altered when Ciproxin is used concomitantly.
Prolongation of bleeding time has been reported during concomitant administration of ciprofloxacin and oral anti-coagulants.
Animal data have shown that high doses of quinolones in combination with some non-steroidal anti-inflammatory drugs, (e.g. fenbufen, but not acetylsalicylic acid) can lead to convulsions.
Transient increases in serum creatinine have been seen following concomitant administration of ciprofloxacin and cyclosporin. Therefore, monitoring of serum creatinine levels is advisable.
The simultaneous administration of quinolones and glibenclamide can on occasion potentiate the effect of glibenclamide resulting in hypoglycaemia.
Renal tubular transport of methotrexate may be inhibited by concomitant administration of ciprofloxacin potentially leading to increased plasma levels of methotrexate. This may increase the risk of methotrexate associated toxic reactions. Therefore, patients receiving methotrexate therapy should be carefully monitored when concomitant ciprofloxacin therapy is indicated.
Concomitant use with probenecid reduces the renal clearance of ciprofloxacin, resulting in increased quinolone plasma levels.

Overdosage, symptoms and treatment
Based on the limited information available in two cases of ingestion of over 18g of ciprofloxacin, reversible renal toxicity has occurred. Therefore, apart from routine emergency measures, it is recommended to monitor renal function, including urinary pH and acidify, if required, to prevent crystalluria. Patients must be kept well hydrated, and in the case of renal damage resulting in prolonged oliguria, dialysis should be initiated.

Serum levels of ciprofloxacin are reduced by dialysis.

Presentation
This infusion is available in 50ml vial (100mg solution for infusion) and 100ml vial (200mg solution for infusion.
One vial is packed into one paper box.

Storage conditions
Protect from light and store below 30°Celsius. Keep out of reach of children.

Shelf-life
The injections can be used within 36 months from the date of manufacture if kept as recommended.

Registration number
CIPROFLO 2MG/ML SOLUTION FOR IV INFUSION - MAL08021506A

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Date of revision of package insert
June 2013